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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

Practice Paper 2 Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-2-Practice-2-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Activity 2: Allotment

2 (a) An allotment is a piece of land that people can rent to grow fruit, flowers and vegetables.

A residents' association has an allotment so that people can grow their own vegetables.

This is a plan of the allotment:

Q2(a) Worked Solution:

318.495 (m2) or 318.5 or 318

Correct total area of rectangular elements

Correct area of triangle or trapezium (m2)

318.495 or 318.5 or 318

5

m2

7.5 m

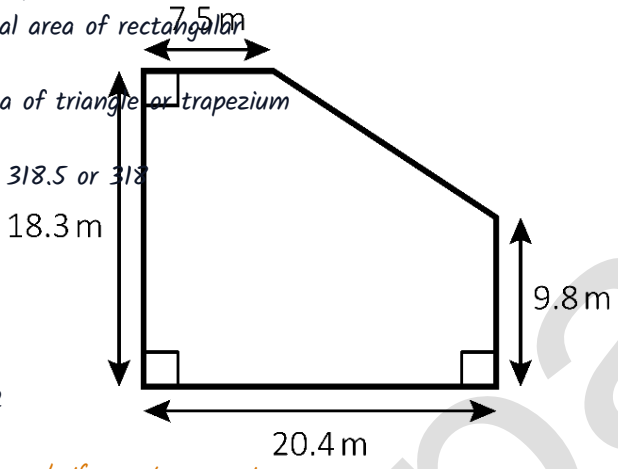
9.8 m

20.4 m

118.575 m2

[3 marks]

Tip: Award 3 marks if correct answer given



Not drawn accurately

What is the area of the allotment?

[3 marks]

Your answer: **m²**

past paper

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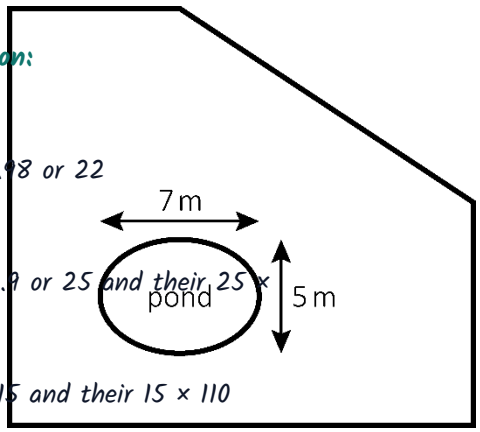
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2 (b) There is an oval pond in the allotment.

The pond is 7 m long, 5 m wide and 80 cm deep.

The pond is to be filled in with soil.

Not drawn accurately



Q2(b) Worked Solution:
(£)1650
7 5
3.14 × × × 0.8 or 21.98 or 22
2 2
21.98 or 22 (m3)
Bag A: their 22 ÷ 0.9 or 25 and their 25 ×
74 or 1850
AND
Bag B: 22 ÷ 1.5 or 15 and their 15 × 110
or 1650
[4 marks]

The formula to find the volume (in m^3) of the pond is: $\pi \times \frac{L}{2} \times \frac{W}{2} \times D$

Tip: Award 4 marks if correct answer given

Where:

L is the length (in m)
W is the width (in m)
D is the depth (in m)

Use $\pi = 3.14$

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Soil can be bought in two sizes of bag.

Bag of soil	Volume of soil	Price per bag
A	0.9 m^3	£74
B	1.5 m^3	£110

The pond should be filled as cheaply as possible.

How much will it cost? Show your working.

[4 marks]

Your answer: **£**

past paper

Please turn over

Examiner use only

2 (c) If 3 people work together, it would take them 8 hours to fill in the pond.

How long would it take 5 people working at the same rate?

[2 marks]

Q2(c) Worked Solution:

4.8 hours or 4 hours and 48 minutes

3 × 8 or 24

4.8 hours or 4 hours 48 minutes

[2 marks]

Tip: Award 2 marks if correct answer given

Your answer:

2 (d) Asif wants to grow potatoes.

The **GrowPotatoes** website gives these instructions for planting potatoes:

> Potatoes should be planted in rows.
>
> There must be 75 cm between the rows, and 50 cm between each plant in a row.
>
> There should be a gap of at least 50 cm between the plants and the edge of the plot.

Q2(d) Worked Solution:

See below (worked example)

4 crosses at corners of usable area

The space for planting potatoes is a rectangle measuring 4 m by 2.5 m

7 columns and 3 rows

Use the scale plan on the next page to show how the potato plants should be arranged to fit in the most potato plants.

Draw a cross (**+**) to mark the position for each plant.

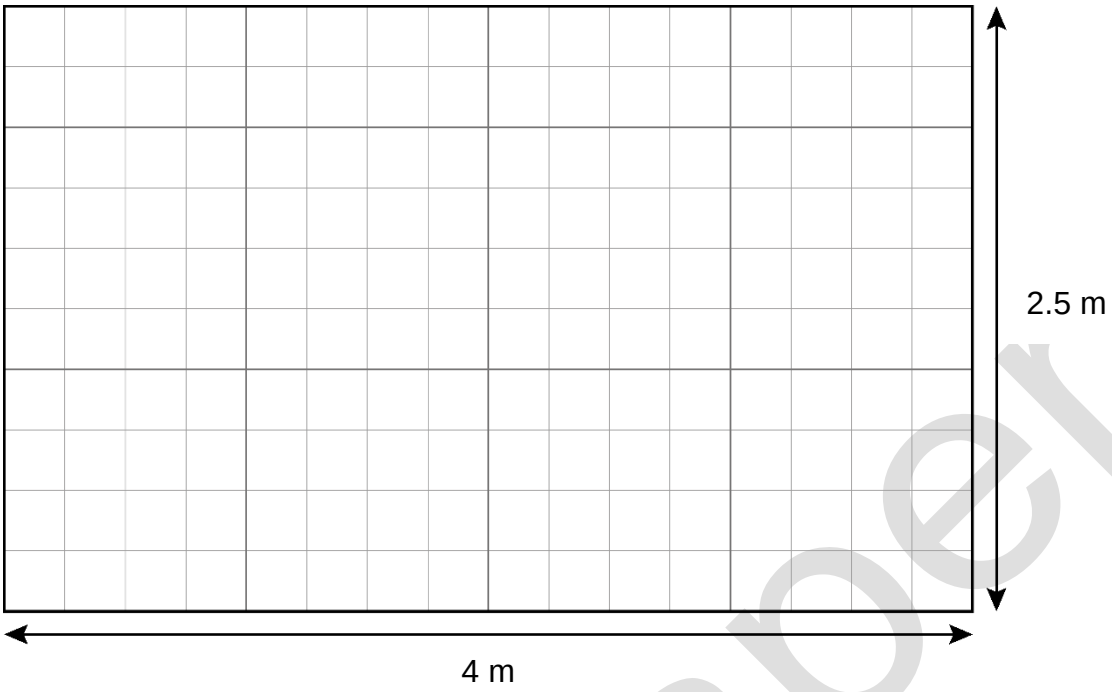
[2 marks]

[2 marks]

Tip: Accept 5 × 4

paper

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Examiner use only

2 (e) Asif finds some trial results about two types of potato, Cara and Desiree.

	Yield (per potato plant)	
Trial number	**Cara**	**Desiree**
1	1.65 kg	1.25 kg
2	1.42 kg	1.42 kg
3	1.88 kg	2.10 kg
4	1.73 kg	1.76 kg
5	1.97 kg	

Yield is the weight of potatoes that can be produced from one potato plant.

Asif wants to plant the type of potato with the higher yield.
He will grow 20 potato plants.

Q2(e) Worked Solution:

Use the trial data to recommend which type of potato Asif should plant and estimate the expected total weight of potatoes, in pounds, that he will get.

Use the conversion: 1 kg = 2.2 pounds

[4 marks]

76.12 (pounds) with Cara
Calculation of correct mean or median for both varieties. Allow rounding to 1dp
Cara
8
20 × their 1.73 or 34.6 (kg)
Correct answer to their 34.6 × 2.2 or 76.12 (pounds)
juice (Calculator Test)
[4 marks]
Tip: Award 4 marks if correct answer given

Your answer:

Type of potato	
pounds	

[Total marks: 15]

past paper

Examiner use only

Activity 3: Fruit juice

3 (a) Pete is a catering student.
He has been given a project testing fruit juicers.

Pete is testing the SuperJuicer and the Blitz-It juicer.

Pete juices 100 g of strawberries in the Blitz-It juicer.
He repeats this 7 times.

Q3(a) Worked Solution:
70 (ml)
[1 mark]

Test number	Volume of juice
1	73 ml
2	66 ml
3	70 ml
4	65 ml
5	67 ml
6	71 ml
7	72 ml

Calculate the median volume of strawberry juice produced by the Blitz-It juicer.

[1 mark]

Your answer: **ml**

past paper

Please turn over

Examiner use only

3 (b) Pete tests the SuperJuicer.

The manufacturer claims that 300 g of apples will produce $\frac{2}{5}$ of a pint of juice.

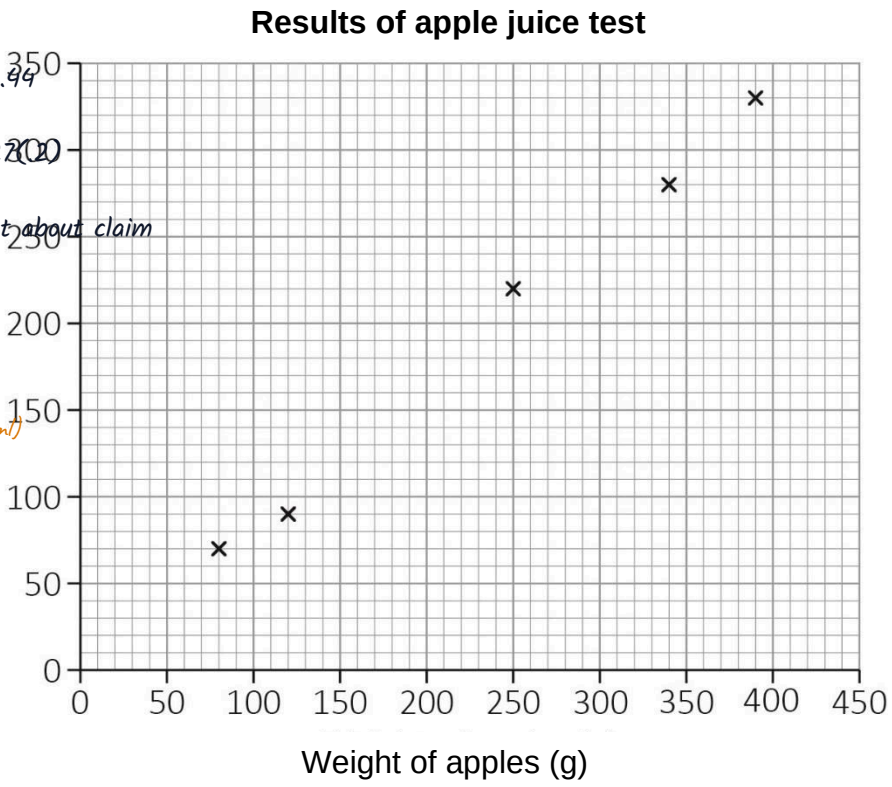
Pete tests this claim by juicing different weights of apples and measuring the amounts of juice they produce.

Q3(b) Worked Solution:

He uses his results to draw this scatter diagram:

250 (ml)
240 to 260 (ml)
their 250 ÷ 568 or 0.44
or
(2 ÷ 5) × 568 or 227(.2)
0.4 or 227(.2)
And correct statement about claim
[3 marks]

Tip: Accept 240 to 260 (ml)



Examiner use only

Is the manufacturer's claim correct? Show how you decide.

Use the conversion: 1 pint = 568 ml

[3 marks]

Your answer:

past paper

Please turn over

Examiner use only

3 (c) The Blitz-It juicer's information states that the weight of juice produced from oranges is around 30% of the weight of the fruit.

If Pete wants to produce 150 g of juice, what weight of oranges should he use?

[2 marks]

Q3(c) Worked Solution:
500 (g)
150 ÷ 0.3 or (150 × 100) ÷ 30
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: **g**

3 (d) Pete asks a group of college students which machine's juice they prefer.

This table shows their responses:

Age group	SuperJuicer	Blitz-It juicer
Under 19 years old	16	4
19+ years old	17	13

Q3(d) Worked Solution:
4
50

A student from the group is chosen at random.

What is the probability that the student is under 19 and preferred juice from Blitz-It juicer?

[2 marks]

Denominator of 50
2
or or 0.08 or 8%
50 25
[2 marks]
Tip: Accept rounded answer: 0.4, 0.41, 0.410

Your answer:

past paper

Examiner use only

3 (e) Pete makes a smoothie using juice from pineapples and mangoes.

He wants the volume of pineapple juice to mango juice to be as close as possible to a 3 : 2 ratio.

Typical volumes of juice produced from 1 fruit are:

Q3(e) Worked Solution:

Pineapple	0.2 litres
Mango	0.08 litres

(£)62(.00)

Costs of fruit at the local supermarket are:

Type of fruit	Price for 1 fruit
Pineapple	£1.50
Mango	65p

(Pineapple) 4.8 (litres)

What is the cost of the fruit that Pete needs to buy to make 8 litres of smoothie?

(Mango) 3.2 (litres)

[5 marks]

9
their 4.8 ÷ 0.2 or 24
and their 3.2 ÷ 0.08 or 40
(Pineapple) 24 and (Mango) 40
(their (24) × 1.5) + (their (40) × 0.65)
or 62
[5 marks]
Tip: Award 5 marks if correct answer given

Your answer: £

Please turn over

past paper

Examiner use only

3 (f) When Pete goes to buy the fruit, he finds that pineapples have increased in price to £1.75 each.

By what percentage has the price of pineapples increased?
Give your answer to 1 decimal place.

[2 marks]

Q3(f) ~~Worked Solution:~~
16.7(%)
(1.75 – 1.5) ÷ 1.5 or 0.166... or 16.66...
in the theatre (Calculator Test)
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: **%**

[Total marks: 15]

past paper

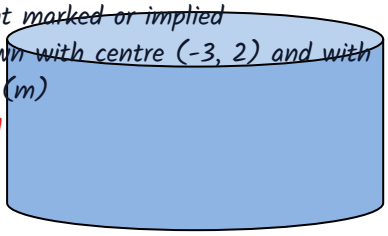
Examiner use only

Activity 4: Working in the theatre

4 (a) Jan is an apprentice in a theatre.
She needs to plan the layout of the stage for the next play.

A prop is an object used on stage.
One of the props is a cylinder with radius 1.5 m

Q4(a) Worked Solution:
-3, 2 point marked or implied
Circle drawn with centre (-3, 2) and with radius 1.5 (m)
[2 marks]



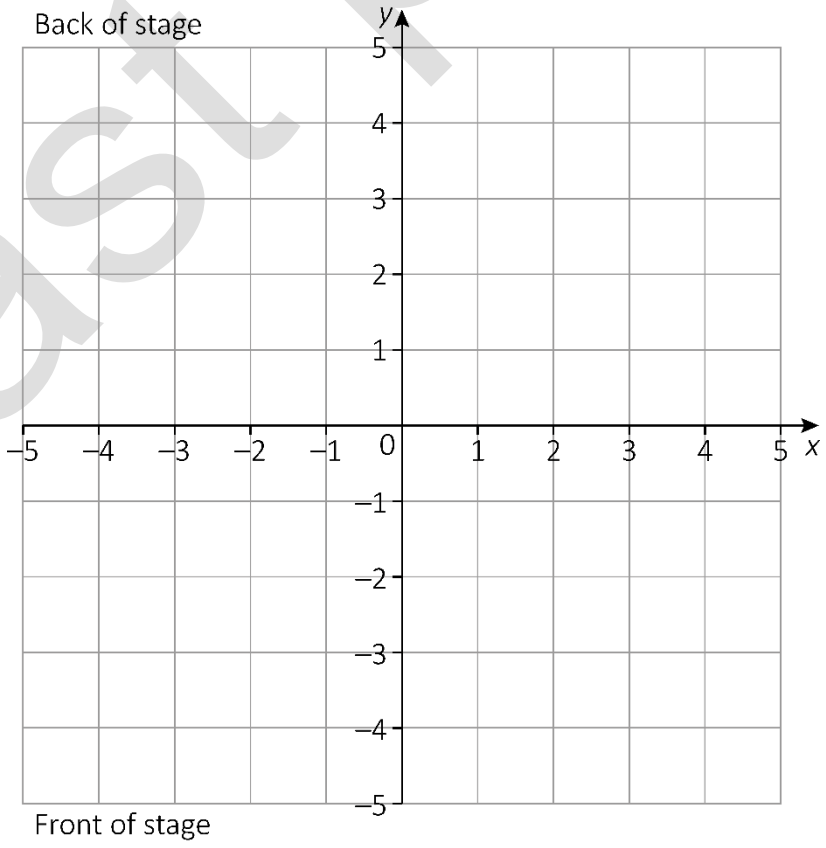
Jan uses a coordinate grid to plan where to put props on the stage.

Each small square on the grid represents a 1 m by 1 m square on the real stage.

She wants the centre of the cylinder to be at (–3, 2)

On the grid, draw a plan view of the cylinder in this position.

[2 marks]



Please turn over

past paper

Examiner use only

4 (b) The cylinder has radius 1.5 m and height 1.25 m

Calculate the volume of the cylinder in m^3
Give your answer to 1 decimal place.

Use $\pi = 3.14$

[2 marks]

Q4(b) ~~Worked Solution~~:
8.8 (m3)
3.14 × 1.52 × 1.25
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: **m^3**

past paper

Examiner use only

4 (c) Jan needs to cover a ball with glitter.



The ball's radius is 60 cm

The formula to find the surface area (A) of a sphere is: $A = 4\ \pi r^2$

Jan buys the glitter in bottles.

She needs to use 300 g of glitter per square metre.

Each bottle holds 254 g of glitter.

Calculate the number of bottles of glitter needed to cover the sphere.
Show your working.

Use $\pi = 3.14$

[4 marks]

Q4(c) Worked Solution:

4 × 3.14 × 0.6 × 0.6
or
4 × 3.14 × 60 × 60
4.5(216) or 45216
their 4.5(216) × 300 or 1356.48
or their 45216 ÷ 10000 × 300
6 (bottles)
[4 marks]
Tip: Accept rounding to at least 2 sig fig.

Your answer: **bottles**

past paper

Please turn over

Examiner use only

4 (d) The theatre director wants a spotlight that will give a circle of light on the stage with a diameter of approximately 2.15 m

The formula for working out the diameter of the light on the stage is:

$$D = \frac{B \times H \times 18}{1000}$$

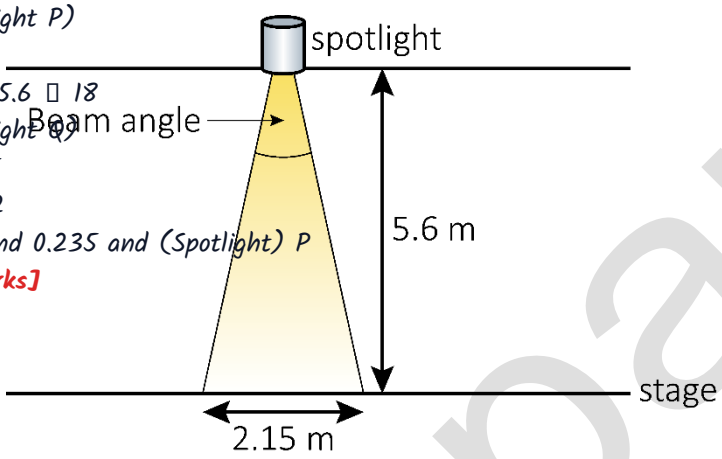
Q4(d) Worked Solution:

10
1

Where:

D is the diameter of the circle of light (in m)
B is the beam angle (in degrees)
H is the vertical distance from the spotlight to the stage (in m)

19 □ 5.6 □ 18
(Spotlight P)
1000
26 □ 5.6 □ 18
(Spotlight Q)
P 1.915
Q 2.62
0.47 and 0.235 and (Spotlight) P
[3 marks]



past paper

Examiner use only

This table shows the available spotlights:

Spotlight	Beam angle
P	19°
Q	26°

Which spotlight should be chosen to get the diameter closest to 2.15 m?
Explain how you decide.

[3 marks]

Your answer:

past paper

Examiner use only

4 (e) The theatre needs new stage curtains.
Jan has been asked to order the fabric for the new curtains.

The total width of fabric needs to be $3\frac{1}{5}$ times the width of the stage front.

Write $3\frac{1}{5}$ as a decimal number.

[1 mark]

Q4(e) Worked Solution:

3.2

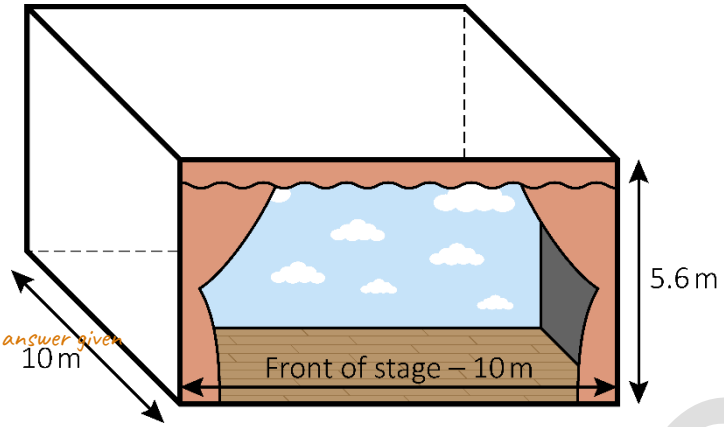
[1 mark]

Your answer:

past paper

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4 (f) This is a diagram of the stage:



Q4(f) Worked Solution:

133.4 (m)

1

10 × 35 or 32

their 32 ÷ 1.4 or 22.857

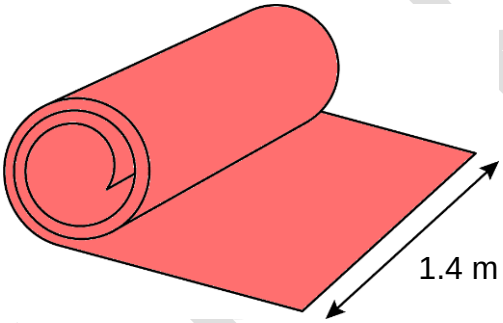
11

[3 marks]

Tip: Award 3 marks if correct answer given

The total width of fabric needs to be $3\frac{1}{5}$ the width of the stage front.

Curtain fabric is sold in widths of 1.4 m



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Several widths of fabric are sewn together to make the curtains.

The length of the curtains needs to be 5.8 m to allow for hems.

What length of fabric does she need?

[3 marks]

Your answer: **m**

[Total marks: 15]

This is the end of the assessment.

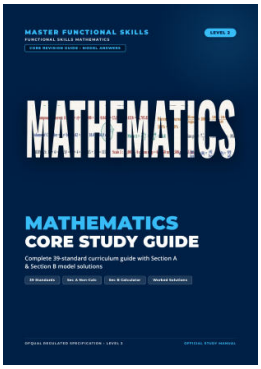
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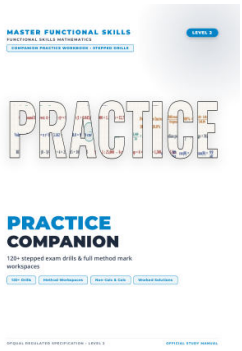


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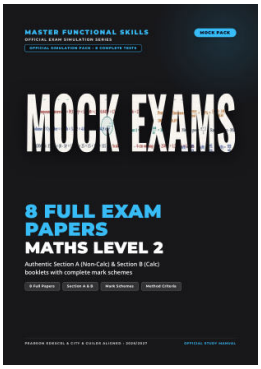


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